

KSP Mun Robotic Mission 2.0

The goal of the mission is to enter Mun orbit from a free-return trajectory with a spacecraft composed of a lander and a command module, separate the lander for a high-precision landing, and perform a surface-to-orbit ascent to execute a successful rendezvous with the orbiting command module before returning to Kerbin to land in a defined area with a 60km radius.

i Initial Conditions:

- Space vehicle composed of a command module and a 2-stages lander
- Start at 5:52 on day 6 already in a Mun flyby trajectory
- Initial speed of 2723 m/s
- Initial cap of 103°

i Constraints:

- a** - Stay at least 30 minutes on the surface to perform science
- b** - Insure a safe moon landing in daylight without risk of sun dazzle:
 - b₁** - Sun elevation = 30°
 - b₂** - Sun relative azimuth < 90
- c** - Minimize the number of critical maneuvers
- d** - Have a free return trajectory
- e** - Minimize cost and DeltaV
- f** - Minimize the duration of the mission
- g** - Return with only one atmospheric entry

1 Mun Capture & Orbit:

The goal here is to capture into a circular orbit around the Mun. In order to do so, a ΔV for circularization is required. To find its value, this next formula is needed.

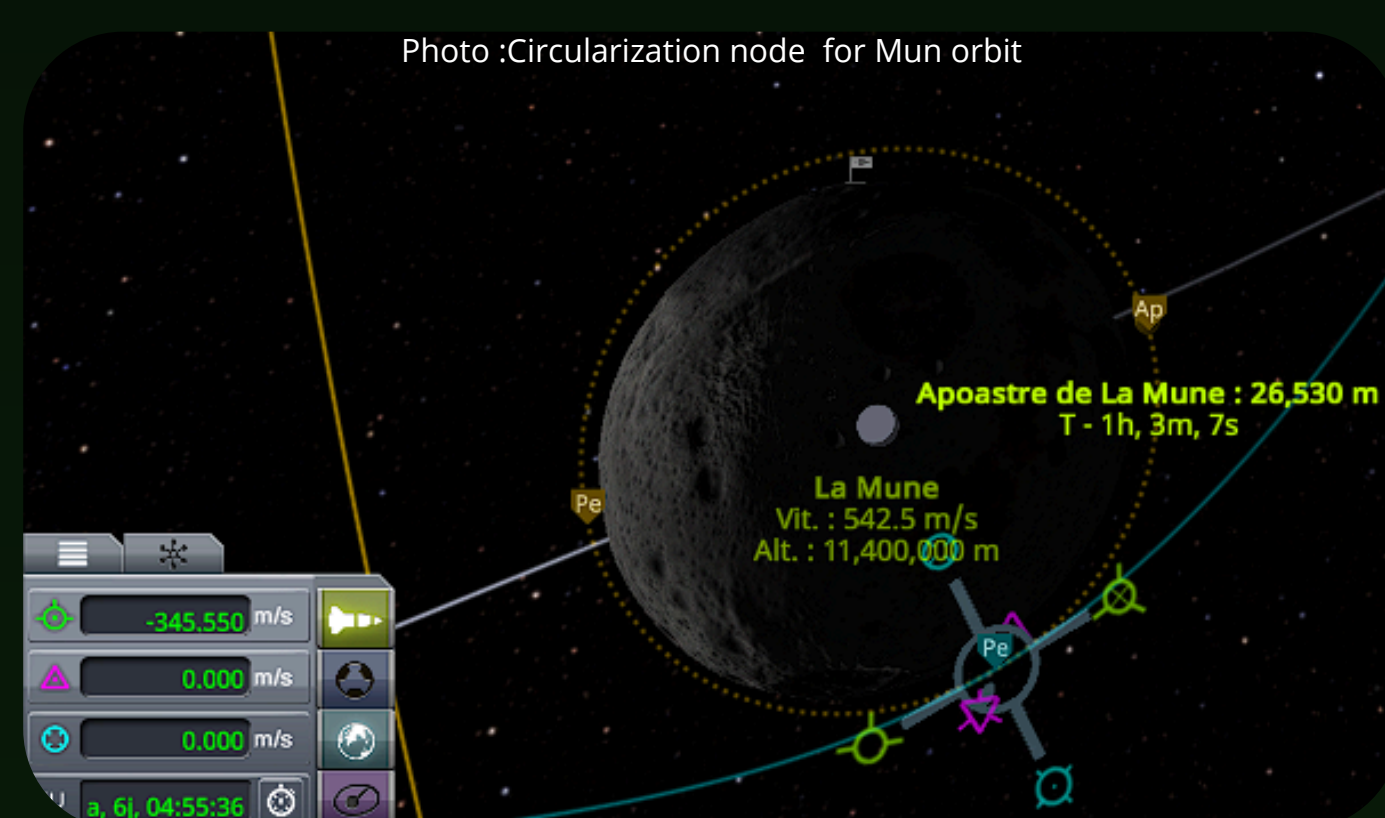
$$\Delta V_{circu} = \sqrt{\Delta V_{in}^2 + \frac{2\mu}{r_p} - \frac{2\mu}{r_{soi}}} - \sqrt{\frac{\mu}{r_p}}$$

When the space vehicle enter the Sphere Of Influence (SOI) of the Mun, its speed is equal to 505.7 m/s. So, $\Delta V_{in} = 505.7$.

The periapsis altitude from the current flyby is 25796 m. This give us the periapsis radius which is the periapsis altitude add to the mun radius, $R_p = 225.8$ km.

The SOI radius is equal to 2,429.2 km.
The gravitational parameter equals 6.5138e10 m³/s².

Finally, when all of these parameters are insert into the formula, the result is that $\Delta V_{circu} = 345.55$ m/s.



2 Mun Landing:

a Sunlight Conditions:

The mission success depends on satisfying specific lighting constraints at the landing site: a solar elevation α_s of approximately 30° and a Solar Relative Azimuth (SRA) below 90° to ensure terrain visibility and prevent pilot dazzle during the final approach.

1. Solar Elevation Modeling (α_s)

The solar elevation is a function of the site's latitude (ϕ) and the Local Hour Angle (h):
 $\sin(\alpha_s) = \cos(\phi) \cdot \cos(h)$

2. Relative Azimuth (SRA) and Operational Safety

The SRA indicates the sun's position relative to the landing trajectory (assuming a standard Westward approach):

- Lunar Morning ($h < 0$): The sun is in the East (SRA > 90°). This is the Safe Zone (Green), where the sun is behind the lander, illuminating the cratered relief without glare.
- Lunar Afternoon ($h > 0$): The sun is in the West (SRA > 90°). This is the Danger Zone (Red), posing a high risk of sun dazzle for the crew or optical sensors.

3. Calculating the Local Hour Angle (h)

It is possible to do as if the Mun were in place of Kerbin because the distance Mun/Kerbin is negligible compared to the distance Kerbin/Sun.

The Local Hour Angle defines the angular offset between the landing site's meridian and the Sun's position in an inertial frame.

- Solar Angular Position (θ_{sun}): Driven by Kerbin's orbital motion:
 - $\theta_{sun} = \text{kerbin} \cdot ut$ (ut = universal time in seconds)
- Site Angular Position (θ_{site}): Combines the Mun's initial orientation, its sidereal rotation, and the target longitude (λ_{site}):
 - $\theta_{site} = (\text{LSIDMun}, t=0) + (n_{mun} \cdot ut) + \lambda_{site}$

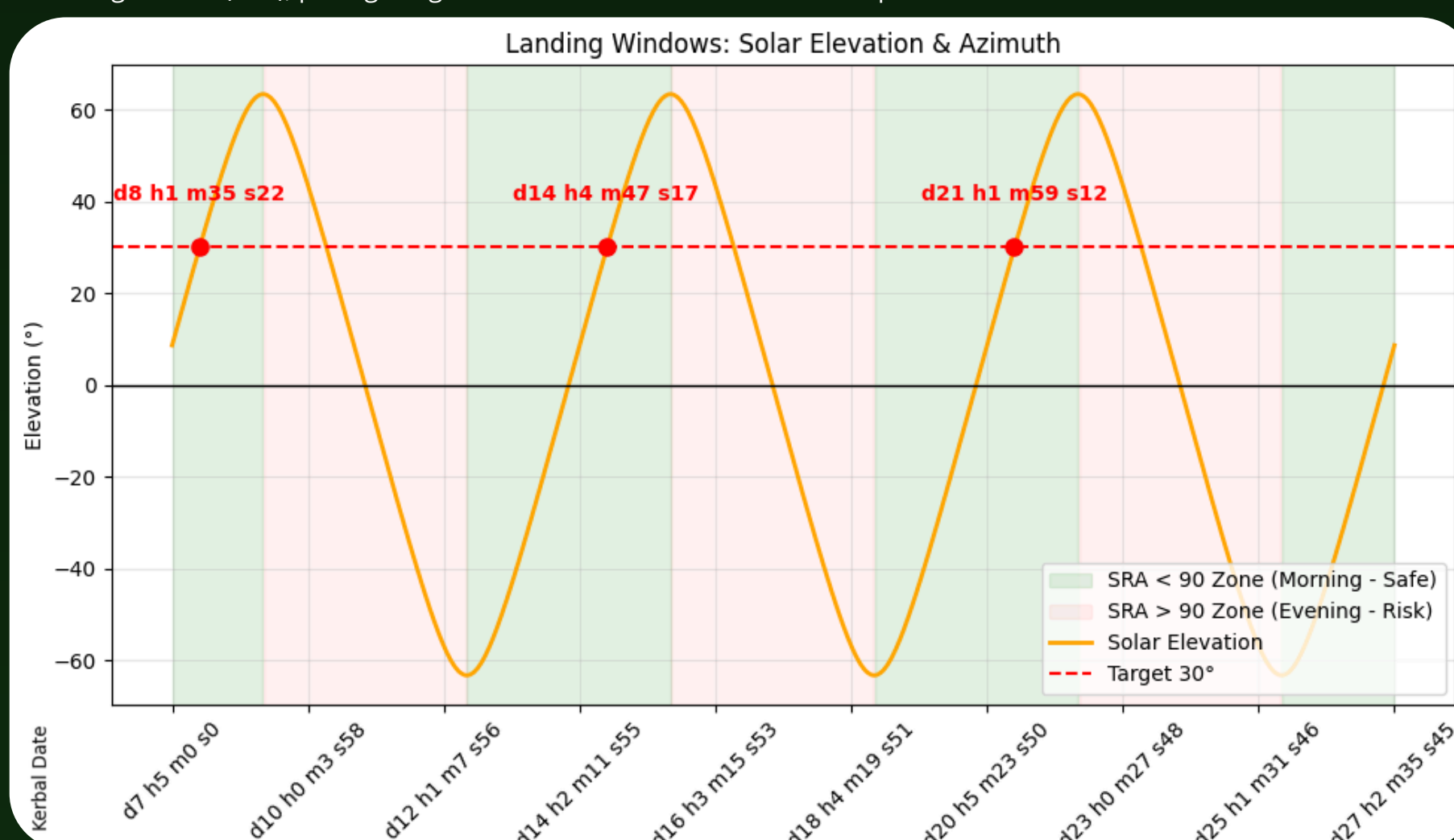
Local Hour Angle (h):

- $h = \text{normalize}(\theta_{site} - \theta_{sun})$
- $h=0$: Solar Noon (Sun at zenith).
- $h < 0$: Morning window.

4. Mission Constants & Inputs

- Target Site: 26.6789°N latitude, 33.0919°E longitude.
- Mun Reference: Initial rotation (LSID0) of 230°.
- Time Scale: Kerbin sidereal year of 9,190,800 seconds.
- Mean anomaly Kerbin at $t=0$: 3.14 rad

Given the time for one orbit around the Mun at 26000m, it is needed to wait 2 orbits before landing on day 8 at approximately 1h.



b Landing sequence:

The landing sequence is divided into three phases:

- Deorbit Maneuver (I→B): At the initial point (I), a retrograde burn is performed to lower the periapsis (r_B) to a target altitude of 6,000 m above the surface.
 - Target Periapsis: $r_B = R_{Mun} + 6,000$ m.
 - Arrival Velocity (v_B) →

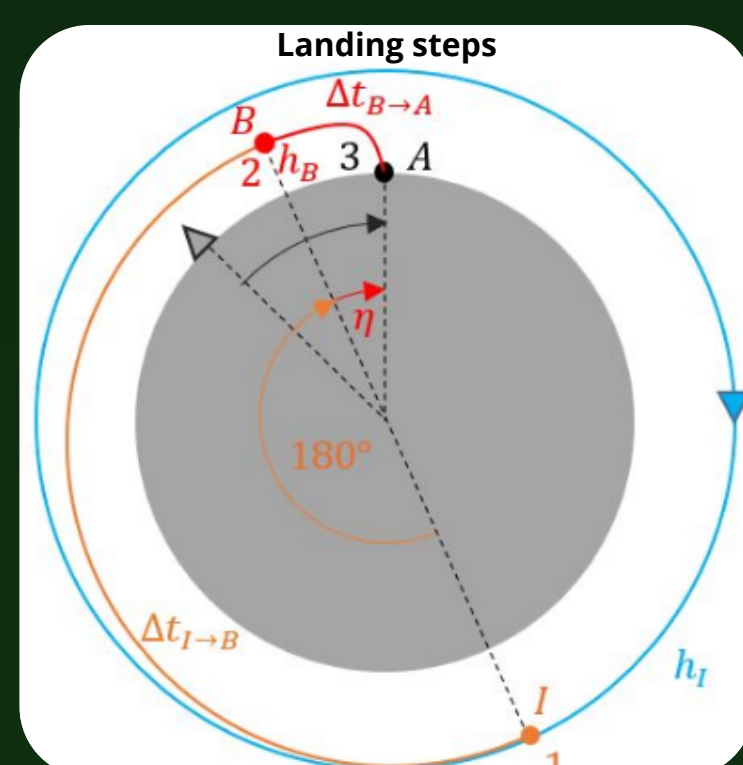
$$v_B = \sqrt{2\mu_{Mun} \left(\frac{1}{r_B} - \frac{1}{r_I} \right)}$$

- Powered Descent (B→A): At point B, the engine is ignited to cancel horizontal velocity. The angle η is required to time this ignition precisely relative to the landing site (A) and is calculated with the Downrange Distance between A and B.

$$D_{BA} = \frac{2v_B^2}{g_{Mun}} \cdot \left(\frac{TWR}{4TWR^2 - 1} \right) \quad \eta = \frac{D_{BA}}{R_{Mun}} \cdot \frac{180}{\pi} \approx 6.7^\circ$$

Final Execution: The total angular offset between the deorbit burn (I) and the landing site (A) is $\Phi_{IA} = 180^\circ + \eta = 186.7^\circ$.

By setting the landing site flag as a target in KSP, the maneuver nodes can be precisely placed using the relative phase angle to ensure the suicide burn terminates exactly at the target coordinates.



- Sluicide burn: To finalize the descent and ensure a soft touch-down, a vertical braking maneuver, known as a Suicide Burn, is executed once horizontal velocity is nullified. This technique maximizes efficiency by delaying the final ignition until the last possible moment.
 - After the horizontal braking phase, the lander enters a vertical free-fall from an altitude (h_0) of 2,000 m.
 - Ignition Altitude Calculation: Using the Thrust-to-Weight Ratio (TWR) of 5.48, the precise ignition altitude ($h_{ignition}$) is determined to cancel vertical velocity exactly at the surface:

$$h_{ignition} = \frac{h_0}{TWR} = \frac{2,000}{5.48} \approx 365 \text{ m}$$

The engines are ignited at full thrust at 365m, resulting in a soft landing at the target site.

As calculated, the lander land on Day 8 at 0h 45min, we are in the defined sunlight window.

3 Ascent & Rendezvous:

To optimize the orbital rendezvous maneuver for time and fuel efficiency, the lander initiates its ascent at point A and enters an intermediate green orbit at point B. While the theoretical ideal for minimizing time would be to start the Hohmann transfer immediately (B=C), a safety margin (t_{margin}) is required to account for the physical duration of the maneuvers. The transfer begins at point C, targeting a rendezvous with the command module in the blue orbit at point E. While an optimized theoretical rendezvous would occur at point D at a specific date (datemodD) estimated via KSP, the actual rendezvous at point E incorporates the necessary timing adjustments to ensure a safe and precise docking.

Here is the equation made to find the moment where the module need to take off:

$$\text{date_take_off} = \text{date_modD} - (t_{AB} + t_{BC} + t_{CE}) + t_{margin}$$

To account for the "human factor" specifically the unpredictable time required to execute the take off and the maneuvers the intermediate green orbit serves as a strategic buffer. Because the lander travels at a higher orbital velocity than the command module while in this lower orbit, it naturally "catches up" to its target. This phase allows to precisely adjust the timing for the Hohmann transfer at point C, providing the flexibility to wait for the optimal phase angle. By incorporating this waiting period in the safety margin (t_{margin}), the mission compensates for manual execution delays, ensuring that the final rendezvous at point E remains accurate despite the imprecision.

$t_{AB} = 180s$ (Observed on KSP), $t_{BC} = 0s$ (not realistic so included in the margin), $t_{margin} = 90s$.

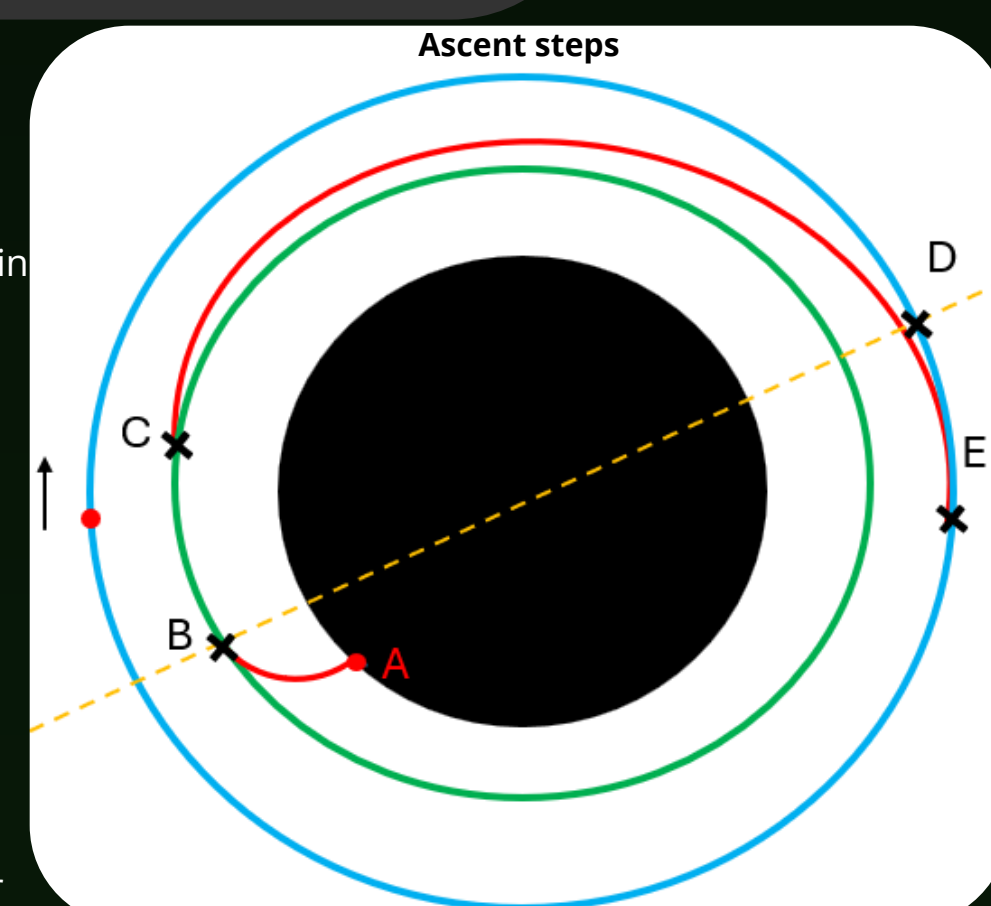
t_{CE} is calculated as half an orbit with a perigee at 14km (green orbit) and an apogee at 26km (blue).

$$T_{CE} = \frac{1}{2} T = \frac{1}{2} \cdot 2\pi \sqrt{\frac{a^3}{\mu}} \quad a = \frac{r_{perigee}(B) + r_{perigee}(C)}{2}$$

We obtain: $t_{CE} = 1269.30$ seconds

By observation in KSP (giving an angle of 5° between A and B):
date_modD= d8 h1 m50 s46

After calculation, takeoff should occur at: d8 h1 m28 s6, aiming for an initial parabolic trajectory that reach an apoapsis at 14000 m altitude.



4 Return to kerbin:

a Find departure time:

Objective:

Land at maximum 60km from 4° N; 68° 30' W with only one atmospheric entry.

To achieve this precision return, we must synchronize the departure from the Mun with the rotation of Kerbin so that the landing site aligns with the arrival trajectory.

Key Parameters & Constraints:

- Kerbin Rotation Period = 21,549 s
- Mun Orbital Period=138,984 s
- Trajectory Duration: The return trip from the Mun's hidden face (departure point) to Kerbin's vacuum perigee (35 km) takes approximately **18,000 s**.
- Atmospheric Shift: After reaching the vacuum perigee at 35 km, the lander travels an additional ($\Delta \lambda_{atm} \approx 13.8^\circ$) longitude arc through the atmosphere before touching down. This shift is accounted for in the targeting.

Trajectory Synchronization Strategy:

We depart from the hidden face of the Mun (Point C) to target a periapsis at 35 km altitude on the hidden face of Kerbin (Point B). We track two positions over time:

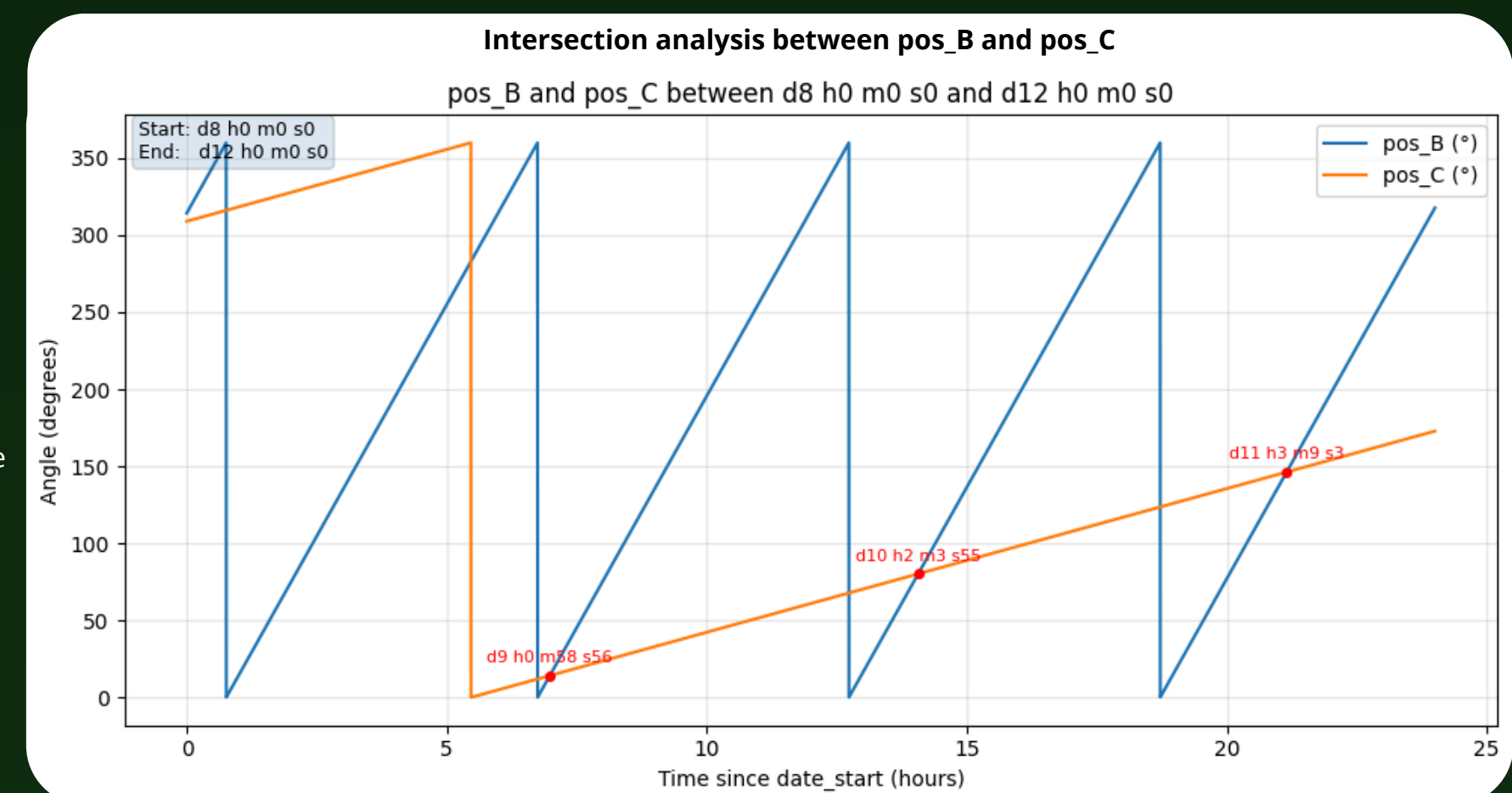
- 'pos_B(t) : The longitude of the target landing site relative to the arrival point (adjusted for Kerbin's rotation during the 18,000s transit).
- 'pos_C(t) : The longitude of the Mun's hidden face (optimal departure point).

A valid launch window occurs when $\text{'pos_B(t)} \equiv \text{'pos_C(t)} \pmod{360}^\circ$.

$$\text{'pos}_B(t) = \lambda_{target} - \Delta \lambda_{atm} + \text{LSid}_0_{Kerbin} + \omega_{Kerbin} \cdot (t + \Delta t_{transit})$$

Where:

- $\lambda_{target} = -68.5^\circ$: Target Longitude (West is negative).
- $\Delta \lambda_{atm} = 13.8^\circ$: Longitude shift due to atmospheric drag (Earth rotation during descent from 35km).
- $\text{LSid}_0_{Kerbin} = 90^\circ$: Initial Sidereal Rotation of Kerbin at $t = 0$.
- $\omega_{Kerbin} = \frac{360}{21549}$: Kerbin's rotation rate ($\approx 0.0167^\circ/s$).
- t : Departure time (JT seconds).
- $\Delta t_{transit} = 18000s$: Duration of return transfer.



b Trajectory Phasing and Correction:

The intersection analysis (above) identifies a potential departure window at Day 9, 0h 58m. However, there is a phasing mismatch: at this exact time, the spacecraft is not correctly positioned at the Mun's hidden face to perform the burn.

However the optimal position is reached 3,120 s before this date. So we just have to extend the last orbit's period to exactly 3,120 s, this requires raising the apoapsis to 78,816 m, after this orbit, the spacecraft will arrive at the optimal position exactly on Day 9, 0h 58m.

c Landing precision analysis:

To quantify the precision of this landing, we calculate the great-circle distance between the actual landing site and the target coordinates using the Haversine formula. This formula provides the shortest distance between two points on a sphere, essential for navigation on planetary surfaces.

$$d = 2R \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta \phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta \lambda}{2} \right)} \right)$$

With ϕ being the latitudes in radians, λ being the longitudes in radians and R being the radius of Kerbin (600 km).

After calculation, the distance between the landed site and the target is equal to 55587 m which is under the limit of 60 km.

This mission is a success because the lander landed not too far from the target and the lander was not damaged, meaning that the kerbals were safe.

